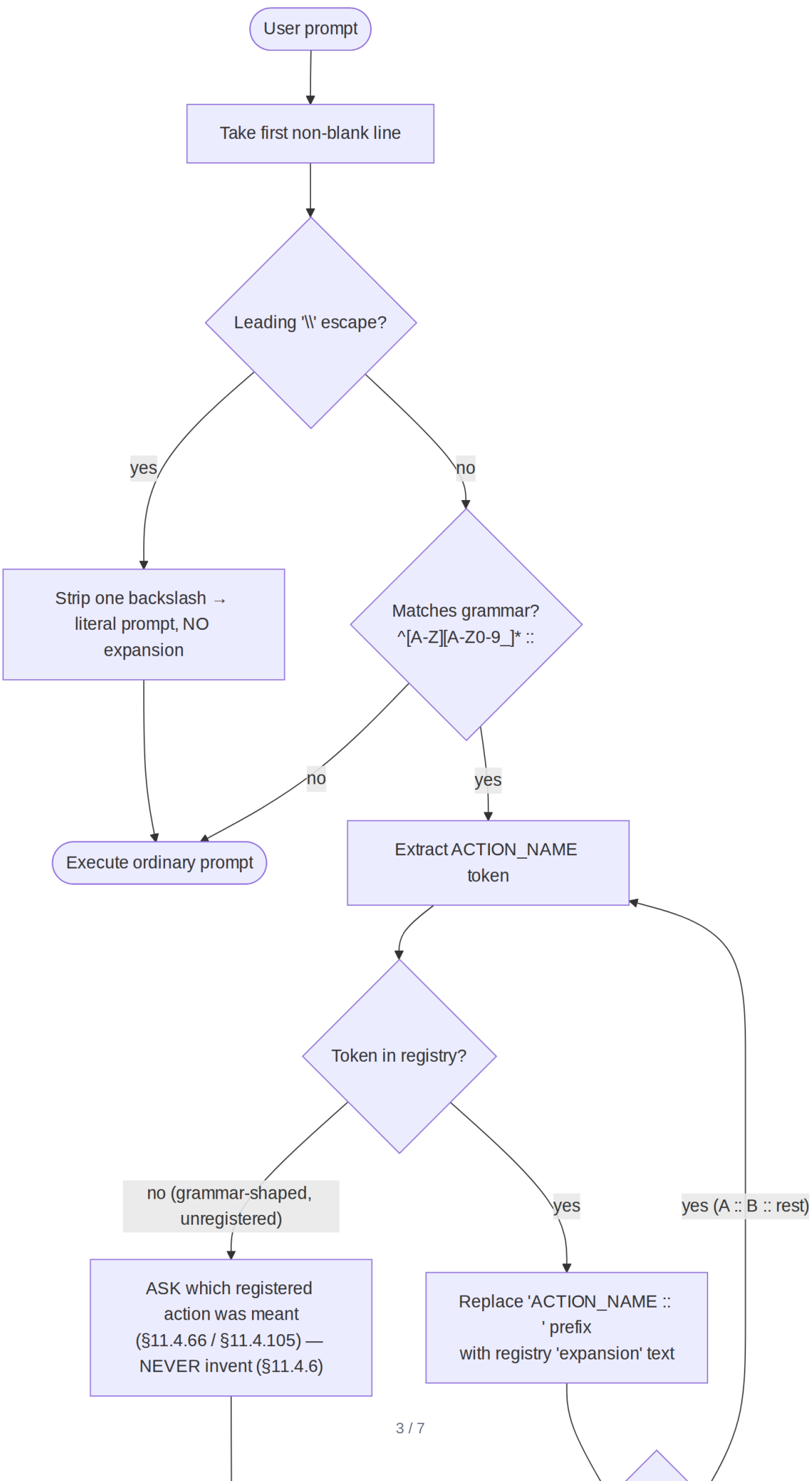
Action-Prefix System — Diagrams

Field	Value
Revision	2
Created	2026-06-09
Last modified	2026-07-02T00:00:00Z
Status	active
Scope	Mermaid diagrams for the §11.4.140 action-prefix system
Audience	maintainers + engineers (companion to USER_GUIDE.md + MANUAL.md)
Inputs	`docs/research/action_prefix_system/{DESIGN.md,GRAMMAR_ADDENDUM.md,IMPLEMENTATION_REPORT.md}`

> Anti-bluff (§11.4.6): the diagrams depict only the design + the verified implementation. All five forms — including the `DEFAULT::` namespace forms 2/4 and the arrow form 5 (`--->`) — are implemented and verified (grammar suite 166/0, paired-mutation meta-test 6/0); none remain spec-only.

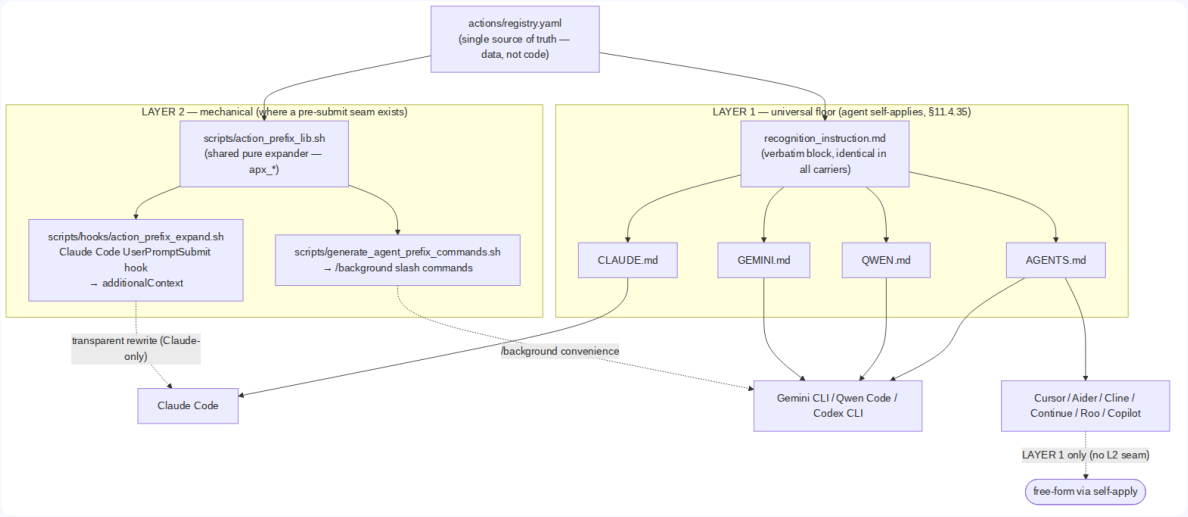
. Expansion flow (prompt → prefix match → registry lookup → expand → execute)

How a single prompt is processed: detect the prefix on the first non-blank line, handle the escape, look the token up in the registry, replace the prefix with the expansion, then execute the residual task. This is the C-preprocessor expand-then-rescan model applied to prompts (DESIGN.md §4).


User prompt
Take first non-blank line
Leading '\\' escape?
yes
no
Strip one backslash → literal prompt, NO expansion
Matches grammar? ^[A-Z][A-Z0-9_]* ::
no
yes
Execute ordinary prompt
Extract ACTION_NAME token
Token in registry?
no (grammar-shaped, unregistered)
yes
yes (A :: B :: rest)
ASK which registered action was meant (§11.4.66 / §11.4.105) — NEVER invent (§11.4.6)
Replace 'ACTION_NAME :: ' prefix with registry 'expansion' text

. Two-layer architecture across CLI agents

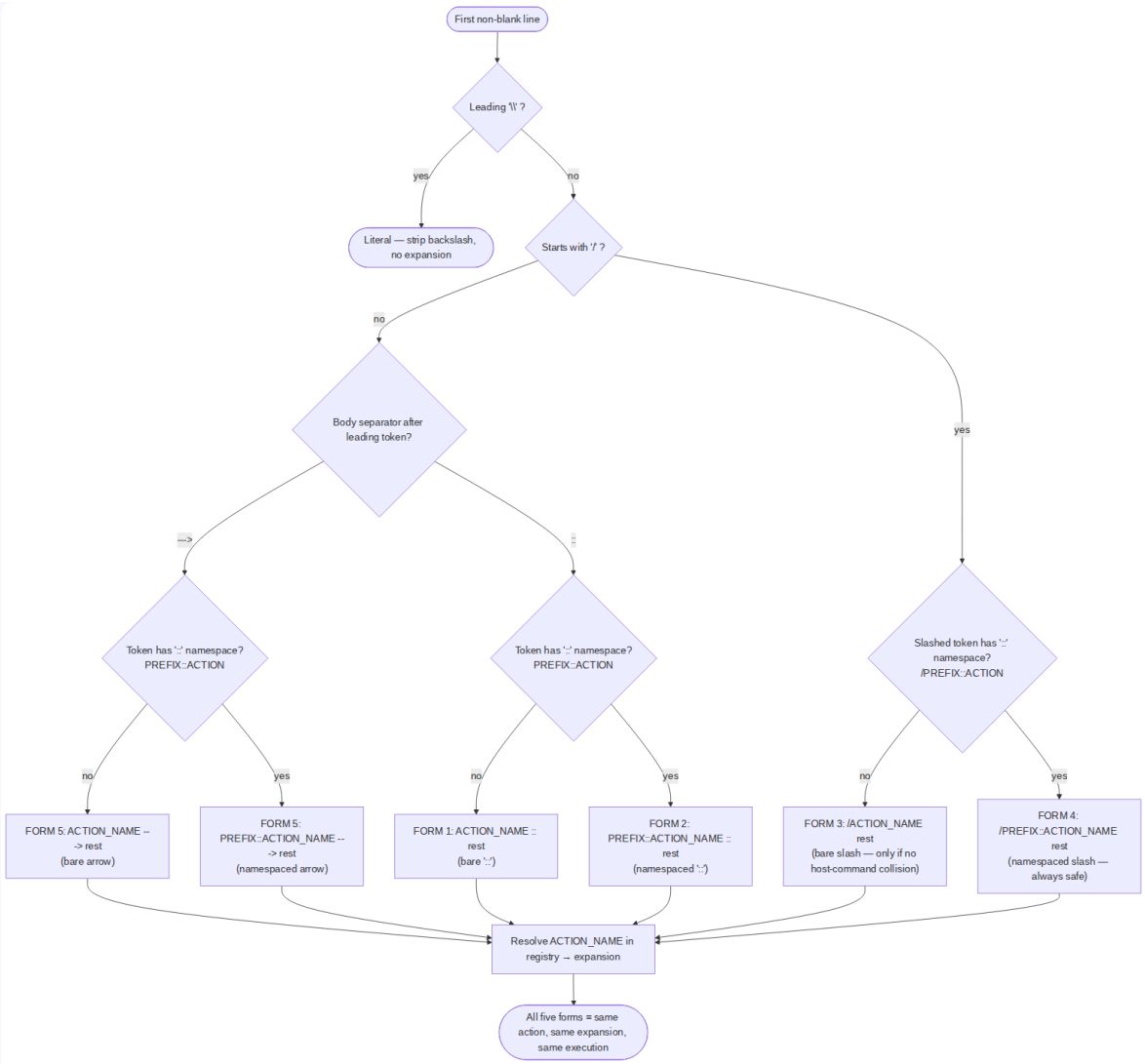
One shared registry feeds both layers. LAYER 1 (the recognition instruction in every context carrier) is the universal floor; LAYER 2 is the mechanical upgrade where an agent exposes a pre-submit seam. Only Claude Code has transparent mechanical free-form interception (the honest §11.4.3 boundary).



3 5

. The -form grammar decision tree

Which form of the first non-blank line was used. All five forms are **implemented** and verified (grammar suite 166/0, paired-mutation meta-test 6/0) per the GRAMMAR_ADDENDUM; forms 2 and 4 (the `DEFAULT::` namespace) and form 5 (the arrow `--->`) are no longer spec-only. All five resolve to the same action + expansion by design.



> Legend: all nodes/edges = implemented + verified (forms 1, 2, 3, 4, 5; grammar suite 166/0, paired-mutation meta-test 6/0). The `DEFAULT::` namespace registry keys (`default_namespace` / `namespace_separator` / `slash_prefix` / `namespaces` / `slash_bare`) plus the arrow keys (`arrow_form_regex` / `arrow_body_separator`) are present and `apx_parse_prefix` returns `matched_form ∈ {colon, slash, arrow}` for all five forms. A mid-line `--->` after a `::` token parses as the colon form (the arrow branch falls through). See MANUAL.md §4.3 / §7.

. Add-a-new-action sequence

Adding an action is a one-row registry edit plus a generator re-run — no code change in either layer. (DESIGN.md §9 / MANUAL.md §6.)

> **NOTE — diagram 4 could not be auto-rendered to an image.** The data is shown below as a table for readability. (Source-level mermaid render failed; see the source `.md` for the original diagram definition.)

```
    actor Dev as Maintainer
    participant Reg as actions/registry.yaml
    participant Gen as generate_agent_prefix_commands.sh
    participant Out as actions/generated/*
    participant Lib as action_prefix_lib.sh
    participant Test as unit test + §1.1 mutation
    participant Doc as docs (§11.4.44 / §11.4.65)

    Dev->>Reg: add one actions[] row\n(name, version, summary≥6 words
§11.4.91, expansion, rules?, composes_with?)
    Dev->>Gen: bash scripts/generate_agent_prefix_commands.sh
    Gen->>Lib: apx_validate_registry + apx_lookup_expansion(NEWACTION)
    Lib-->>Gen: expansion (verbatim)
    Gen->>Out: write gemini/newaction.toml, qwen/newaction.toml, codex/
newaction.md\n(/newaction = form 3)
    Note over Reg,Out: No edit to the hook OR the library — registry is
data, not code.\nLAYER 1 already knows to consult the registry.
    Dev->>Test: add unit test (parses + grammar match + expansion correct)
\n+ paired mutation (corrupt expansion → gate FAILs)
    Dev->>Doc: bump revision header; export .html/.pdf at commit time
(conductor)
    Note over Dev: Optional — inline the expansion in the LAYER-1
block\nONLY if it must work without registry-file access (like
BACKGROUND).
```

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. Related documents

- USER_GUIDE.md — end-user guide.

- MANUAL.md — developer/maintainer manual.
- `docs/research/action_prefix_system/{RESEARCH.md, DESIGN.md, GRAMMAR_ADDENDUM.md, IMPLEMENTATION_REPORT.md, RULE_DRAFT.md}` — source design.
- Canonical authority: `Constitution.md` §11.4.140.